

CompTIA A+ Assignment

Module -1, 2 [Hardware and its components]

[Q-1] What is input device?

➤ An input device is a hardware device used to enter data into a computer.

Examples:

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam

[Q-2] What are output device?

➤ An output device shows the result from the computer.

Examples:

- Monitor
- Printer
- Speaker
- Projector

[Q-3] What is CPU?

- CPU (Central Processing Unit) is called the “brain of the computer.” It processes instructions and controls all operations.

[Q-4] What are the types of CPU?

➤ CPU Type :

- Single Core CPU
- Dual Core CPU
- Quad Core CPU
- Hexa Core CPU
- Octa Core CPU

[Q-5] What do we need to keep the CPU Healthy?

- Clean dust regularly,
Use thermal paste,
Keep cooling fan working,
Avoid overheating,
Use proper power supply.

[Q-6] Do a practical to remove processor and apply thermal paste in it and install it again.

➤ Practical: Remove Processor and Apply Thermal Paste

1. Shut down computer and unplug power.
2. Open cabinet.
3. Remove CPU cooler/fan.
4. Unlock processor socket and remove CPU carefully.

5. Clean old thermal paste.
6. Apply small amount of new thermal paste.
7. Reinstall CPU correctly.
8. Lock socket and install cooler again.
9. Start the system.

[Q-7] Do a practical to Identify CPU and its Sockets.

➤ Practical: Identify CPU and CPU Socket

1. Open computer cabinet.
2. Locate processor on motherboard.
3. Check socket type written near CPU or in BIOS/software.
4. Examples of sockets: LGA1151, AM4.

[Q-8] What is memory?

➤ Memory stores data and instructions for the computer.

[Q-9] What are the types of memory?

➤ Primary Memory

- RAM
- ROM

➤ Secondary Memory

- Hard Disk
- SSD
- Pen Drive

[Q-10] Do a practical to identify memory types.

➤ Practical: Identify Memory Types

1. Open cabinet.
2. Locate RAM slots.
3. Check RAM label (DDR3, DDR4, DDR5).
4. Identify storage devices like HDD or SSD.

[Q-11] Do a practical to install memories in system.

➤ Practical: Install Memory in System

1. Turn off computer.
2. Open RAM slot clips.
3. Align RAM notch correctly.
4. Press RAM gently until locked.
5. Start computer and check detection.

[Q-12] Do a practical to identify main memory frequencies.

➤ Practical: Identify Main Memory Frequencies

1. Open BIOS or use software like CPU-Z.
2. Check RAM frequency.
3. Example: 2400 MHz, 3200 MHz

[Q-13] What is bios?

- BIOS (Basic Input Output System) is firmware stored on motherboard. It starts the computer and checks hardware.

[Q-14] Describe working process of BIOS.

➤ Working Process of BIOS

1. Power ON computer.
2. BIOS performs POST (Power On Self Test).
3. Detects hardware devices.
4. Loads bootloader from storage device.
5. Starts operating system.

[Q-15] Do a practical to reset bios when system is on.

➤ Practical: Reset BIOS When System is ON

1. Restart computer.
2. Enter BIOS setup using Del/F2 key.
3. Select “Load Default Settings.”
4. Save and exit.

[Q-16] Do a practical of Hard resetting the BIOS.

➤ Practical: Hard Reset BIOS

1. Turn off computer and unplug power.
2. Remove CMOS battery for 5 minutes.
3. Reinsert battery.
4. Start computer.

[Q-17] Do a practical of identifying BIOS chip from the motherboard □ What is CMOS?

➤ Practical: Identify BIOS Chip

1. Open motherboard cabinet.

2. Locate small chip labeled BIOS/UEFI.
3. Usually near CMOS battery.

[Q-18] What is CMOS?

- CMOS stores BIOS settings like date, time, and boot settings.

[Q-19] What is motherboard?

- Motherboard is the main circuit board connecting all computer parts.

[Q-20] Describe types of motherboards.

- Types of Motherboards

- ATX
- Micro-ATX
- Mini-ITX

[Q-21] Do a practical by identifying parts of motherboard.

- Practical: Identify Parts of Motherboard

- CPU socket
- RAM slots
- PCI slots
- SATA ports
- CMOS battery
- BIOS chip

[Q-22] Do a practical by removing all removable parts from the motherboard.

➤ Practical: Remove Removable Parts from Motherboard

1. Power off system.
2. Remove RAM.
3. Remove CPU fan.
4. Remove storage cables.
5. Remove expansion cards carefully.

[Q-23] What is system bus?

➤ System bus transfers data between CPU, memory, and devices.

[Q-24] What is chipset and types of chipset?

➤ Chipset controls communication between components.

- Types:
 1. Northbridge
 2. Southbridge

[Q-25] Describe how does the Northbridge chipset work?

➤ Working of Northbridge Chipset

- Northbridge connects CPU with:
 1. RAM
 2. Graphics card
 3. High-speed devices

[Q-26] what is SMPS? And its purpose?

- SMPS (Switched Mode Power Supply) converts AC power into DC power for computer parts.

[Q-27] DO a practical to install SMPS.

- Practical: Install SMPS
 1. Open cabinet.
 2. Place SMPS in cabinet slot.
 3. Tighten screws.
 4. Connect motherboard and device power cables.

[Q-28] How to check smps?

- Check fan rotation
Use SMPS tester or multimeter
Check if system powers ON

[Q-29] List out the types of storage devices.

- Types of Storage Devices
 - HDD
 - SSD
 - Pen Drive
 - CD/DVD
 - Memory Card

[Q-30] Describe the working process of storage devices.

- Storage devices save data permanently and provide data when needed.

[Q-31] Do a practical to Remove storage devices and reinstall it and make a gpt disk.

- Practical: Remove and Reinstall Storage Device and Make GPT Disk
 1. Remove SATA/power cable.
 2. Remove storage drive.
 3. Reinstall drive and connect cables.
 4. Open Disk Management.
 5. Convert disk to GPT format.

[Q-32] What is SATA?

- SATA (Serial ATA) is an interface used to connect storage devices.

[Q-33] Describe the working of SATA.

- SATA transfers data between motherboard and storage devices.

[Q-34] Do a practical to install SATA.

- Practical: Install SATA
 1. Connect SATA data cable to motherboard.
 2. Connect SATA power cable from SMPS.
 3. Start computer and check detection.

[Q-35] What is SCSI storage and type of SCSI?

- SCSI (Small Computer System Interface) is a storage interface used in servers.
 - Types:
 1. Parallel SCSI
 2. Serial Attached SCSI (SAS)

[Q-36] What is I/O ports?

- I/O ports connect external devices to computer.

[Q-37] List out the I/O ports available Do a practical to identify the I/O ports?

- List of I/O Ports
 - USB
 - HDMI
 - VGA
 - Ethernet
 - Audio port
 - PS/2 port
- Practical:
 - Identify ports on motherboard or cabinet.

[Q-38] What is Boot Process?

- Boot process is the process of starting the computer and loading the operating system.

[Q-39] Describe the boot process in Linux?

➤ Boot Process in Linux

1. BIOS/UEFI starts system
2. GRUB bootloader loads
3. Linux kernel loads
4. System services start
5. Login screen appears

[Q-40] List out the types of display?

➤ Types of Display

- CRT
- LCD
- LED
- OLED

[Q-41] What is printer? And type of printer

➤ Printer is an output device used to print documents.

- Types:
 1. Inkjet Printer
 2. Laser Printer
 3. Dot Matrix Printer

[Q-42] Do a practical to install the printer

➤ Practical: Install Printer

1. Connect printer to computer.
2. Install printer driver.

3. Add printer in settings.
4. Print test page.

[Q-43] Do a practical to Troubleshoot the improper printing.

➤ Practical: Troubleshoot Improper Printing

1. Check paper and ink/toner.
2. Restart printer.
3. Check cable connection.
4. Reinstall printer driver.
5. Clean print head if needed.

[Q-44] What are the parts of laptop?

➤ Parts:

- Display
- Keyboard
- Touchpad
- Battery
- Motherboard
- RAM
- Storage
- Cooling fan

[Q-45] Do a practical to disassemble the laptop.

➤ Practical: Disassemble Laptop

1. Turn off laptop and remove charger.
2. Remove battery if removable.
3. Open screws carefully.

4. Remove back cover.
5. Remove RAM, storage, fan, etc. carefully.
6. Reassemble properly.